US Patent & Trademark Office
Patent Public Search | Text View

United States Patent Application Publication	20250277489
Kind Code	A1
Publication Date	September 04, 2025
Inventor(s)	BUSCH; Christian

POSITIVE-DISPLACEMENT MACHINE ACCORDING TO THE SPIRAL PRINCIPLE

Abstract

The invention relates to a positive-displacement machine according to the spiral principle, in particular a scroll compressor, said displacement machine comprising a housing which has a compressor housing and a drive housing, an orbiting displacement spiral and a counter spiral being located in the compressor housing and engaging into one another in such a way that variable compression chambers are formed between the displacement spiral and the counter spiral in order to receive and compress a working medium flowing through a working medium circuit, a drive shaft being located in the drive housing and being drivingly connected to the displacement spiral, a bearing plate being located between the drive housing and the compressor housing and carrying a bearing for the drive shaft, and an anti-rotation mechanism being provided between the bearing plate and the orbiting displacement spiral. The bearing plate is located completely inside the housing, the bearing plate and the bearing being made of materials having coefficients of thermal expansion which differ from one another by at most 20%, in particular by at most 10%, in particular by at most 5%.

Inventors:	**BUSCH; Christian (Feldkirch, AT)**
Applicant:	**OET GMBH** (Lustenau, AT)
Family ID:	**86185057**
Appl. No.:	**18/862737**
Filed (or PCT Filed):	**April 11, 2023**
PCT No.:	**PCT/EP2023/059414**

Foreign Application Priority Data

DE	10 2022 111 377.9	May. 06, 2022

Publication Classification

Int. Cl.: **F04C18/02** (20060101)

U.S. Cl.:

CPC **F04C18/0207** (20130101); F04C2240/30 (20130101); F04C2240/50 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a National Stage Application of International Application No. PCT/EP2023/059414, filed on Apr. 11, 2023, which claims benefit of priority to German Patent Application No. 10 2022 111 377.9, filed on May 6, 2022 which applications are incorporated herein by reference in their entirety. To the extent appropriate, a claim of priority is made to each of the above disclosed applications.

TECHNICAL FIELD
[0002] The invention relates to a positive-displacement machine according to the spiral principle. Such a positive-displacement machine is known, for example, from DE 10 2016 118 525 B4.
[0003] The above-mentioned DE 10 2016 118 525 B4 describes a positive-displacement machine according to the spiral principle, in particular said displacement machine, comprising a scroll compressor with a housing that is made of aluminium for reasons of corrosion protection. The housing forms a wall that separates a drive chamber of the scroll compressor from a compression chamber. The wall carries a drive bearing that supports a drive shaft. The drive shaft eccentrically engages into an orbiting displacement spiral, which, in turn, is spirally engaged with a counter spiral. An orbiting movement of the displacement spiral, which is initiated by the eccentric connection with the drive shaft, creates variable compression chambers between the displacement spiral and the counter spiral, which can be used to compress a working medium.
[0004] The well-known spiral compressor also comprises an anti-rotation mechanism. The anti-rotation mechanism comprises a pin that engages an opening in the bottom of the displacement spiral, thereby preventing the displacement spiral from rotating. In this way, the displacement spiral is thrust into the desired orbital motion.
[0005] DE 10 2016 118 525 B4 has recognized that it is favourable if the pins of the anti-rotation mechanism are not inserted directly into the aluminium wall, but rather a sliding plate made of steel is formed between the aluminium wall and the displacement spiral. The pins of the anti-rotation mechanism are inserted into this sliding plate and engage into openings of the displacement spiral. For reasons of durability, the pins are also made of one steel.
[0006] The wall that carries the sliding plate is simultaneously a bearing plate that supports the bearing for the drive shaft. To ensure good durability, drive shaft bearings are usually made of steel.
[0007] Due to the different materials between the bearing plate and the bearing, there is different thermal expansion in the operation of the positive-displacement machine. This must be taken into account when pressing the bearing into the bearing plate and can lead to the bearing clearance being very small in partial load operation of the positive-displacement machine or to the bearing being preloaded during operation. Both have a negative effect on the service life of the bearing.

Description

[0008] The object of the invention is to specify a positive-displacement machine according to the spiral principle, which comprises an improved service life. In particular, the bearing of the drive shaft is said to comprise an increased service life as a result of the invention.
[0009] Specifically, the invention is according to the idea of specifying a positive-displacement machine according to the spiral principle, in particular a scroll compressor, with a housing comprising a compressor housing and a drive housing. An orbiting displacement spiral and a counter spiral are located in the compressor housing, which engage into one another in such a way that variable compression chambers are formed between the displacement spiral and the counter spiral in order to receive and compress a working medium flowing through a working medium circuit. A drive shaft is located in the drive housing, which is drivingly connected to the displacement spiral, wherein a bearing plate is located between the drive housing and the compressor housing, which carries a bearing for the drive shaft. An anti-rotation mechanism is provided between the bearing plate and the orbiting displacement spiral. According to the invention, the bearing plate is located completely within the housing, the bearing plate and the bearing being made of materials whose coefficient of thermal expansion differs from one another by at most 20%, in particular by at most 10%, in particular by at most 5%.
[0010] The invention essentially makes use of two effects. On the one hand, the choice of material for the bearing plate and bearing ensures that the bearing does not tension with the bearing plate during operation so that the service life of the bearing is increased. On the other hand, the bearing plate is completely located inside the housing. As a result, a different material can be chosen for the bearing plate than for the housing. For example, the housing can be made of aluminium and offer appropriate corrosion protection. The bearing plate, on the other hand, can be made of steel and comprise sufficient strength to hold the bearing. In this respect, the invention is also according to the idea of a functional separation between the housing and the bearing plate. While in the prior art the housing and the bearing plate are often designed in one piece, the invention uses a two-part embodiment so that each component can be optimally configured for the function assigned to it.
[0011] Therefore, by forming the bearing plate and bearing from materials that comprise a similar coefficient of thermal expansion, the press fit between the bearing plate and the bearing can be well dimensioned, which has a positive effect on the service life of the bearing.
[0012] The bearing plate can be connected to the housing in a positive-locking manner. In particular, the bearing plate can be connected to the housing by a press fit, a screw connection, rivets or a pin connection.
[0013] In one embodiment, it is specifically provided that the bearing plate is fixed in a housing ring which is located between the drive housing and the compressor housing. With this variant, the focus is on good ease of maintenance. The housing itself features the drive housing, the housing ring, and the compressor housing. The housing ring is preferably connected to the bearing plate in a positive-locking manner. The material of the housing ring preferably corresponds to the materials of the drive housing and the compressor housing. The bearing plate is preferably made of another material, in particular, steel.
[0014] Specifically, in the case of a preferred embodiment, it is envisaged that the housing ring comprises a different material than the bearing plate and/or the compressor housing. The material of the housing ring is preferably corrosion-resistant, in particular aluminium.
[0015] Alternatively, the bearing plate can be fixed in the drive housing. With such a variant, a separate housing ring is dispensed with. The variety of components is reduced, which has a positive effect on the costs of production.
[0016] Another preferred embodiment provides that the bearing plate is fixed in the compressor housing.
[0017] Another alternative variant of the invention provides that the bearing plate is fixed, in particular in a positive-locking manner, connected to the counter spiral. The above-mentioned

positive-locking connections can be used for all variants.
[0018] In another preferred embodiment of the invention, it is provided that the orbiting displacement spiral comprises a displacement spiral floor in which a sealing element is attached. The element sealing seals the orbiting displacement spiral to the bearing plate and thus to the drive housing. This increases the efficiency of the positive-displacement machine.
[0019] A sliding plate may also be placed between the bearing plate and the orbiting displacement spiral, in particular the displacement spiral floor. Preferably, the sliding plate is completely located inside the housing. It is particularly preferable if the sealing element seals against the sliding plate.
[0020] The sliding plate is preferably made of a material that comprises good sliding and sealing properties. This reduces friction within the positive-displacement machine and increases its efficiency. Simultaneously, which is important for applications in the field of electromobility, vibrations and noise emissions are reduced.
[0021] The invention is explained in more detail below by means of an exemplary embodiment with reference to the attached drawing. In it, the only FIGURE shows a partial cross-sectional view of a positive-displacement machine according to the invention according to a preferred exemplary embodiment.
[0022] FIG. **1** shows a section of a positive-displacement machine, in particular a scroll compressor, with a housing **100**. The housing **100** is divided into a drive housing **10** and a compressor housing **20**. In the embodiment shown here, a housing ring **31** is located between the drive housing **10** and the compressor housing **20**. The housing ring **31** is firmly connected to the drive housing **10** and the compressor housing **20**. Preferably, the drive housing **10**, the compressor housing **20** and the housing ring **31** are made of aluminium alloy.
[0023] In the drive housing **10** there is also a drive shaft **12**. Drive housing **10** can also comprise an electric motor, which is not shown here, which directly drives the drive shaft **12**.
[0024] The drive shaft **12** penetrates a bearing plate **30** and is supported within the bearing plate **30** in a bearing **32**. Bearing **32** is preferably fixed in bearing plate **30** with a press fit. Upstream of bearing **32** is a shaft seal **11**, which encloses the drive shaft **12** and thus closes the drive housing **10** tightly opposite the bearing plate **30**.
[0025] The drive shaft **12** engages with an eccentric bearing **14** via a compensating mechanism **13**. The eccentric bearing **14** is engaged with an orbiting displacement spiral **21**, so that drive energy can be transferred to the orbiting displacement spiral **21** via the drive shaft **12** and the eccentric bearing **14**.
[0026] In the case of the positive-displacement machine in accordance with the FIGURE, an anti-rotation mechanism **40** is also provided. The anti-rotation mechanism **40** comprises a pin **41** firmly fixed in the bearing plate **30**. The pin **41** engages with play in a ring **42**, which is located in an opening of the displacement spiral **21**. Through the play between the ring **42** and the pin **41**, the displacement spiral **21** is forced into orbiting motion by the eccentricity on the drive shaft **12**. A simple rotation of the displacement spiral is thereby prevented.
[0027] In order to ensure a good seal between the orbiting displacement spiral **21** and the bearing plate **30**, the bearing plate **30** also carries a sliding plate **33**. The orbiting displacement spiral **21** also comprises a sealing element **24**, which is pressed against the sliding plate **33** and thus ensures good tightness. The sealing element **24** can be a circumferential O-ring.
[0028] As can also be clearly seen in the FIGURE, the bearing plate **30** is located completely inside the housing. The bearing plate **30** is firmly connected to the counter spiral **25**. In particular, the connection can be established in a positive-locking manner. For example, mounting pins inserted with a press fit can hold the bearing plate **30** to the counter spiral **25**. Alternatively, a screw connection, riveting or a tolerance ring design is also possible.
[0029] The bearing plate **30** can be made of a steel or a cast material. Preferably, a material is used that is also used for the production of the shaft bearing **32**. In any case, the bearing plate **30** and the shaft bearing **32** shall be made of materials which comprise the same or at least similar coefficients

of thermal expansion. In the operation of the positive-displacement machine, the shaft bearing **32** and the bearing plate **30** expand evenly. The press fit between the shaft bearing **32** and the bearing plate **30** can thus be dimensioned well and precisely, which has a positive effect on the service life of the shaft bearing **32**.

REFERENCE LIST

[0030] **10** drive housing [0031] **11** shaft seal [0032] **12** drive shaft [0033] **13** balancing mechanism [0034] **14** eccentric bearing [0035] **20** compressor housing [0036] **21** orbiting displacement spiral [0037] **22** displacement spiral floor [0038] **23** compression chamber [0039] **24** sealing element [0040] **25** counter spiral [0041] **30** bearing plate [0042] **31** housing ring [0043] **32** bearing [0044] **33** sliding plate [0045] **40** anti-rotation mechanism [0046] **41** pin [0047] **42** ring [0048] **100** housing

Claims

1. A positive-displacement machine according to the spiral principle, in particular scroll compressors, comprising a housing comprising a compressor housing and a drive housing, wherein an orbiting displacement spiral and a counter spiral are located in the compressor housing, which engage into one another in such a way that variable compression chambers are formed between the positive-displacement spiral and the counter spiral in order to receive and compress a working medium flowing through a working medium circuit, wherein, in the drive housing, a drive shaft is located, which is drivingly connected to the displacement spiral, wherein, a bearing plate is located between the drive housing and the compressor housing, which bearing plate carries a bearing for the drive shaft, and wherein an anti-rotation mechanism is provided between the bearing plate and the orbiting positive-displacement spiral, wherein the bearing plate is located entirely within the housing, wherein the bearing plate and the bearing are made of materials whose coefficient of thermal expansion differs by at most 20%, in particular by at most 10%, in particular by at most 5%.

2. The positive-displacement machine according to claim 1, wherein the bearing plate is connected to the housing in a positive-locking manner.

3. The positive-displacement machine according to claim 1, wherein the bearing plate is fixed in a housing ring which is located between the drive housing and the compressor housing.

4. The positive-displacement machine according to claim 3, wherein the housing ring comprises a material other than the bearing plate, in particular the same material as the drive housing and/or the compressor housing.

5. The positive-displacement machine according to claim 1, wherein the bearing plate is fixed in the drive housing.

6. The positive-displacement machine according to claim 1, wherein the bearing plate is fixed in the compressor housing.

7. The positive-displacement machine according to claim 1, wherein the bearing plate is firmly connected to the counter spiral, in particular in a positive-locking manner.

8. The positive-displacement machine according to claim 1, wherein the orbiting displacement spiral comprises a displacement spiral floor in which a sealing element is attached.

9. The positive-displacement machine according to claim 1, wherein a sliding plate is located between the bearing plate and the orbiting displacement spiral, in particular the displacement spiral floor.

10. The positive-displacement machine according to claim 9, wherein the sliding plate is located completely within the housing.

11. The positive-displacement machine according to claim 9, wherein the sealing element seals against the sliding plate.